Prioritisation Method for the Synthesis Core

Remote sensing and geospatial approaches for multi-hazard monitoring in Nepal

Transparent, reproducible selection of the 100-article in-depth core from the 256 in-scope articles.

Rationale

256 in-scope articles exceed what can be synthesised in depth in one narrative review. Rather than a blunt cut (e.g., a single year threshold, which would discard the foundational 2015 Gorkha cascade literature), the core was selected with an explicit weighted priority score applied uniformly to every article and computed only from variables already recorded - so the ranking is reproducible and free of per-paper subjectivity.

Priority score

Each in-scope article received a score from 0 to 100, the weighted sum of six normalised criteria reflecting the review's aims:

Criterion	Definition (normalised 0-1 sub-score)	Weight
Multi-hazard relevance	hazard searches the paper appears in (1=0; 2=0.6; 3=0.8; >=4=1.0)	25
Thematic centrality	Nepal study area + RS method + hazard focus (screening relevance)	25
Operational content	capability tier (early-warning 1.0 -> inventory 0.2)	15
Recency	(Year - 2016)/(2025 - 2016)	15
Cross-database validation	indexed in both Scopus and Web of Science	10
Citation impact	percentile of citations per year (age-normalised)	10

Priority = 25(MultiHazard) + 25(Centrality) + 15(Operational) + 15(Recency) + 10(CrossDB) + 10(Citation). Citations are age-normalised (citations per year) so recent work is not penalised; weights are stated explicitly and can be varied for sensitivity analysis.

Selection: stratified by hazard

To preserve cross-hazard balance, the core was selected within hazard strata, each guaranteed a floor of ten articles (or all of its articles where fewer than ten exist - the case for drought, with only seven); remaining places up to 100 were allocated in proportion to stratum size, and the top-scoring articles within each stratum were taken.

Hazard stratum	In-scope (n)	Selected for core
Landslide	113	39
Multi-hazard / cascade	69	23
GLOF / glacier	34	16
Flood	33	15
Drought	7	7 (all)
Total	256	100

Outcome and QA

- 100 articles selected; 80 are from 2020 or later, while foundational pre-2020 studies (incl. 2015 Gorkha cascade work) are retained.
- All 256 in-scope articles remain in the quantitative/bibliometric layer; the core defines only the set read in full for the synthesis and maturity matrix.
- A final manual QA pass replaced five weaker fits (generic deep-learning architectures and pure glacier-process studies) with Nepal-specific hazard studies in the same hazard stratum. Combined with hazard-tag confirmation for records resolved by full-text reading after this ranking was first run, the final core composition is Landslide 39, Multi-hazard 23, GLOF 16, Flood 15, Drought 7 (Figure 2 of the manuscript).

The ranking, sub-scores, and core flag are recorded in the screening workbook and reproduced by rank_priority.py; editing the weights, floor or target re-derives the core transparently.